

ISRAEL JACOB ORTIZ

Machine Learning / AI Engineer • Python • PyTorch • LLMs • MLOps • RAG • Model Deployment

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PROFESSIONAL SUMMARY

Machine learning / AI engineer with 5+ years of professional experience building, training, and deploying production ML and LLM systems in **Python, PyTorch, and TensorFlow**. Hands-on with model fine-tuning, evaluation, MLOps, and LLM applications (RAG, agents) — with rigorous, honest benchmarking and reliable production deployment.

PROFESSIONAL EXPERIENCE

Machine Learning / AI Engineer | *SUCCESS Magazine*

Apr 2023 – Present

ML Models • LLMs • MLOps

Washington, DC (Hybrid)

- **Train and deploy ML models** (PyTorch, scikit-learn) for classification, ranking, and anomaly detection, reducing false positives by about 40% with calibrated scoring.
- **Build LLM applications** — RAG and agentic systems (Claude, OpenAI, LangChain, LangGraph) — integrated into production with evaluation and guardrails.
- **Own MLOps** — training pipelines, model registry, monitoring, and CI/CD (MLflow, SageMaker) — for reliable, reproducible deployment of 10+ models in production.

Founder & Principal ML Engineer | *Ambrose*

Jul 2021 – Present

Boutique AI/ML Consultancy • Part-Time

Remote

- **Founded and run Ambrose**, a boutique AI/ML consultancy, leading a small team that delivers model fine-tuning, RAG products, and LLM agents for startups and enterprise clients on AWS and GCP.

ML Engineer / Team Lead | *SPCA International*

Sep 2020 – Mar 2023

Applied ML — Team Lead

Washington, DC

- **Led a four-engineer ML team** over two and a half years, shipping nine model releases with a disciplined evaluation process.
- **Built the evaluation harness** — representative sampling, held-out benchmarking, and calibration — that gated launches and caught over-optimistic results before release.

Freelance ML / AI Engineer | *Independent / Contract*

Feb 2019 – Jun 2021

Solo Client Work Prior to Founding Ambrose

Remote

- **Delivered model and LLM projects** for early clients as an independent contractor — building the track record that led to founding Ambrose.

TECHNICAL SKILLS

ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost; classification, ranking, anomaly detection, NLP, computer vision, model evaluation

LLMs & GenAI: Anthropic Claude, OpenAI, Hugging Face; fine-tuning, RAG (pgvector, Pinecone), agents (LangChain, LangGraph), prompt engineering

MLOps & Deployment: MLflow, SageMaker, Vertex AI, ONNX; model registry, monitoring, CI/CD, Docker, Kubernetes

Languages & Data: Python (expert), SQL; Pandas, NumPy, Spark; feature engineering, data pipelines, experiment tracking (W&B)

EDUCATION

Virginia Tech — *M.Eng., Computer Science (Machine Learning & Artificial Intelligence)*

2023 – 2024

GPA: 4.0 | Full-Ride & Tech Talent Investment Program (TTIP) Scholar

Relevant coursework: Machine Learning, Deep Learning, Large Language Models, MLOps, Statistical Inference, Computer Vision, NLP, Model Evaluation & Calibration.

University of Mary Washington — *B.S., Computer Science & Mathematics (Double Major)*

2019 – 2022

GPA: 3.8 | Dean's List 7/8 Semesters

Honors & Leadership: FBLA Collegiate Chapter — President & Founder; University-Wide Machine Learning Competition — 1st Place; IT Advisory Board — Treasurer.

CERTIFICATIONS

AWS Certified Machine Learning – Specialty • Google Cloud Professional Machine Learning Engineer • DeepLearning.AI Deep Learning Specialization • Anthropic Claude Developer (Builder Track)